

Assignment no. 2

Due Date 09 April 2026 (11:00 pm)

Covers material:

- i. LS Regression Line
- ii. Standard Error of Estimate
- iii. Testing regression coefficients
- iv. Prediction interval and confidence interval

Involves:

Please complete all work by using **RStudio** and upload the result **RStudio** Console on AI teaching platform.

Question 1.

An educational economist wants to establish the relationship between an individual's income and education. He takes a random sample of 10 individuals and asks for their income (in \$1000s) and education (in years). The results are shown below.

Income	25	33	22	41	18	28	32	24	53	26
Education	11	12	11	15	8	10	11	12	17	11

By using R find:

- a) LS Regression Line for income on education level
- b) Standard Error of Estimate
- c) Do the data provide sufficient evidence at the 5% significance level to conclude that higher education level leads to higher income?
- d) Predict 90% confidence interval for the income when an individual has 16 years of education level.

Question 2.

It is doubtful that any sports collects more statistics than baseball. The fans are always interested in determining which factors lead to successful teams. The table below lists the team batting average and the team winning percentage for the 14 league teams at the end of a recent season.

Team-B-A	Winning%
0.254	0.414
0.269	0.519
0.255	0.500
0.262	0.537
0.254	0.352
0.247	0.519
0.264	0.506
0.271	0.512
0.280	0.586
0.256	0.438
0.248	0.519
0.255	0.512
0.270	0.525
0.257	0.562

$y = \text{winning \%}$ and $x = \text{team batting average}$

By using R find:

a) LS Regression Line

b) Standard Error of Estimate

c) Do the data provide sufficient evidence at the 5% significance level to conclude that higher team batting average lead to higher winning percentage?

d) Predict with 90% confidence interval for the winning percentage of a team whose batting average is 0.275.